

Orthogonalität

$$\textcircled{1} \quad \vec{v}_1 = \begin{pmatrix} -2 \\ 0 \\ 2 \end{pmatrix}, \vec{v}_2 = \begin{pmatrix} 0 \\ 0 \\ 4 \end{pmatrix}, \vec{v}_3 = \begin{pmatrix} 3 \\ -4 \\ -2 \end{pmatrix}$$

$$v_1 = w_1$$

$$w_2 = -\frac{\langle v_2, w_1 \rangle}{\langle w_1, w_1 \rangle} \cdot w_1 + v_2$$

NR:

$$\langle v_2, w_1 \rangle = \begin{pmatrix} 0 \\ 0 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 0 \\ 2 \end{pmatrix} = 8$$

$$\langle w_1, w_1 \rangle = \begin{pmatrix} -2 \\ 0 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 0 \\ 2 \end{pmatrix} = 8$$

$$w_2 = \begin{pmatrix} 2 \\ 0 \\ -2 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 4 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix}$$

$$w_3 = -\frac{\langle v_3, w_1 \rangle}{\langle w_1, w_1 \rangle} \cdot w_1 - \frac{\langle v_3, w_2 \rangle}{\langle w_2, w_2 \rangle} \cdot w_2 + v_3$$

NR:

$$\langle v_3, w_1 \rangle = \begin{pmatrix} 3 \\ -4 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 0 \\ 2 \end{pmatrix} = -6 - 4 = -10$$

$$\langle w_1, w_1 \rangle = 8$$

$$\langle v_3, w_2 \rangle = \begin{pmatrix} 3 \\ -4 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix} = 6 - 4 = 2$$

$$\langle w_2, w_2 \rangle = \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix} = 8$$

$$w_3 = \frac{10}{8} \cdot w_1 - \frac{2}{8} w_2 + v_3$$

$$w_3 = \frac{5}{4} w_1 - \frac{1}{4} w_2 + v_3$$

$$w_3 = \frac{5}{4} \cdot \begin{pmatrix} -2 \\ 0 \\ 2 \end{pmatrix} - \frac{1}{4} \cdot \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix} + \begin{pmatrix} 3 \\ -4 \\ -2 \end{pmatrix}$$

$$w_3 = \begin{pmatrix} -\frac{5}{2} \\ 0 \\ \frac{5}{2} \end{pmatrix} - \begin{pmatrix} \frac{1}{2} \\ 0 \\ \frac{1}{2} \end{pmatrix} + \begin{pmatrix} 3 \\ -4 \\ -2 \end{pmatrix}$$

$$w_3 = \begin{pmatrix} -3 \\ 0 \\ 2 \end{pmatrix} + \begin{pmatrix} 3 \\ -4 \\ -2 \end{pmatrix} = \begin{pmatrix} 0 \\ -4 \\ 0 \end{pmatrix}$$

$$\|w_1\| = \sqrt{2^2 + 2^2} = \sqrt{4+4} = \sqrt{8}$$

$$b_1 = \frac{1}{\sqrt{8}} \cdot \begin{pmatrix} -2 \\ 0 \\ 2 \end{pmatrix}$$

$$\|w_2\| = \sqrt{8}$$

$$b_2 = \frac{1}{\sqrt{8}} \cdot \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix}$$

$$\|w_3\| = \sqrt{4^2} = 4$$

$$b_3 = \frac{1}{4} \cdot \begin{pmatrix} 0 \\ -4 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix}$$

②

$$\langle x, y \rangle = 2x_1y_1 - x_1y_2 - x_2y_1 + x_2y_2$$

$$\langle x, y \rangle = (x_1 \ x_2) \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} \cdot \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \quad ?$$

③

$$f_1(x) = 3x + 1$$

$$f_2(x) = 2x - 2$$

$$\langle f, g \rangle = \int_0^1 f(x)g(x)dx$$

$$f_1 = w_1 \quad \langle w_1, w_1 \rangle = \int_0^1 (3x+1) \cdot (3x+1) dx = \int_0^1 9x^2 + 6x + 1 dx$$

$$= 3x^3 + 3x^2 + x \Big|_0^1 = 7$$

$$\|w_1\| = \sqrt{7}$$

$$b_1 = \frac{1}{\sqrt{7}} \cdot (3x+1)$$

$$w_2 = -\frac{\langle f_2, w_1 \rangle}{\langle w_1, w_1 \rangle} \cdot w_1 + f_2$$

$$\langle f_2, w_1 \rangle = \int_0^1 (2x-2) \cdot (3x+1) dx = \int_0^1 6x^2 - 4x - 2 dx$$

$$= 2x^3 - 2x^2 - 2x \Big|_0^1 = -2$$

$$\langle w_1, w_1 \rangle = 7$$

$$w_2 = \frac{2}{7} \cdot (3x+1) + (2x-2)$$

$$w_2 = \frac{6}{7}x + \frac{2}{7} + \frac{14}{7}x - \frac{14}{7}$$

$$w_2 = \frac{20}{7}x - \frac{12}{7}$$

$$\|w_2\| = \sqrt{\langle w_2, w_2 \rangle}$$

$$\langle w_2, w_2 \rangle = \int_0^1 \left(\frac{20}{7}x - \frac{12}{7} \right) \cdot \left(\frac{20}{7}x - \frac{12}{7} \right) dx =$$

$$\int_0^1 \frac{400}{49}x^2 - \frac{480}{49}x + \frac{144}{49} dx$$

$$= \frac{400}{49} \cdot \frac{1}{3}x^3 - \frac{480}{49} \cdot \frac{1}{2}x^2 + \frac{144}{49}x$$

$$\langle w_2, w_2 \rangle = \frac{16}{21}$$

$$\|w_2\| = \frac{4}{\sqrt{21}}$$

$$b_2 = \frac{\sqrt{21}}{4} \cdot \left(\frac{20}{7}x - \frac{12}{7} \right)$$

$$\textcircled{4} \quad u_1 = w_1$$

$$w_2 = - \frac{\langle u_2, w_1 \rangle}{\langle w_1, w_1 \rangle} \cdot w_1 + u_2$$

NR:

$$\langle u_2, w_1 \rangle = \begin{pmatrix} -2 \\ -1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} -3 \\ 3 \\ 2 \end{pmatrix} = 6 - 3 = 3$$

$$\langle w_1, w_1 \rangle = \begin{pmatrix} -3 \\ 3 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} -3 \\ 3 \\ 2 \end{pmatrix} = 9 + 9 + 4 = 22$$

$$w_2 = -\frac{3}{22} \cdot \begin{pmatrix} -3 \\ 3 \\ 2 \end{pmatrix} + \begin{pmatrix} -3 \\ -1 \\ 0 \end{pmatrix}$$

?

⑥

$$u_1 = \begin{pmatrix} 3 \\ 0 \\ -2 \end{pmatrix} \quad u_2 = \begin{pmatrix} 3 \\ 4 \\ 1 \end{pmatrix}$$

$$\langle v, u_1 \rangle = \left\langle \begin{pmatrix} 3 \\ 0 \\ -2 \end{pmatrix}, \begin{pmatrix} 3 \\ 0 \\ -2 \end{pmatrix} \right\rangle = \frac{6}{3} + \frac{4}{3} = \frac{10}{3}$$

$$\langle v, u_2 \rangle = \left\langle \begin{pmatrix} 3 \\ 0 \\ -2 \end{pmatrix}, \begin{pmatrix} 3 \\ 4 \\ 1 \end{pmatrix} \right\rangle = \frac{6}{3} - \frac{4}{3} = -\frac{1}{3}$$

$$u = \frac{10}{3} \cdot \begin{pmatrix} 3 \\ 0 \\ -2 \end{pmatrix} - \frac{1}{3} \cdot \begin{pmatrix} 3 \\ 4 \\ 1 \end{pmatrix}$$

$$u = \begin{pmatrix} 10 \\ 0 \\ -20 \end{pmatrix} - \begin{pmatrix} 1 \\ 4 \\ 1 \end{pmatrix} = \begin{pmatrix} 9 \\ -4 \\ -21 \end{pmatrix}$$

$$\textcircled{5} \quad u_1 = \begin{pmatrix} 0 \\ -1 \\ 3 \end{pmatrix}, u_2 = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$$

a)

$$w_1 = u_1$$

$$\|w_1\| = \sqrt{10}$$

$$b_1 = \frac{1}{\sqrt{10}} \cdot \begin{pmatrix} 0 \\ -1 \\ 3 \end{pmatrix}$$

$$w_2 = - \frac{\langle u_2, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1 + u_2$$

$$\langle u_2, w_1 \rangle = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ -1 \\ 3 \end{pmatrix} = 1 + 3 = 4$$

$$\langle w_1, w_1 \rangle = \begin{pmatrix} 0 \\ -1 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ -1 \\ 3 \end{pmatrix} = 1 + 9 = 10$$

$$w_2 = -\frac{2}{5} \cdot \begin{pmatrix} 0 \\ -1 \\ 3 \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$$

$$w_2 = \begin{pmatrix} 0 \\ \frac{2}{5} \\ -\frac{6}{5} \end{pmatrix} + \begin{pmatrix} 1 \\ -\frac{5}{5} \\ \frac{5}{5} \end{pmatrix} = \begin{pmatrix} 1 \\ -\frac{3}{5} \\ \frac{1}{5} \end{pmatrix}$$

$$\|w_2\| = \sqrt{1 + \frac{9}{25} + \frac{1}{25}} = \frac{\sqrt{25}}{5}$$

$$b_2 = \frac{5}{\sqrt{25}} \cdot \begin{pmatrix} 1 \\ -\frac{3}{5} \\ \frac{1}{5} \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{25}}{5} \\ -\frac{1}{5} \\ \frac{1}{5} \end{pmatrix}$$

$$b) \quad v = \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix}$$

$$\langle v, b_1 \rangle = \frac{1}{\sqrt{10}} \cdot \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ -1 \\ 3 \end{pmatrix} = \frac{1}{\sqrt{10}} \cdot 6 = \frac{6}{\sqrt{10}}$$

$$\langle v, b_2 \rangle = \frac{5}{\sqrt{35}} \cdot \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} -\frac{1}{5} \\ \frac{2}{5} \\ \frac{1}{5} \end{pmatrix} = \frac{5}{\sqrt{35}} \cdot \left(\frac{10}{5} - \frac{2}{5} \right) = \frac{8}{\sqrt{35}}$$

$$u = \frac{6}{\sqrt{10}} \cdot \frac{1}{\sqrt{10}} \cdot \begin{pmatrix} 0 \\ -1 \\ 3 \end{pmatrix} + \frac{8}{\sqrt{35}} \cdot \frac{5}{\sqrt{35}} \cdot \begin{pmatrix} -\frac{1}{5} \\ \frac{2}{5} \\ \frac{1}{5} \end{pmatrix}$$

$$u = \frac{6}{10} \cdot \begin{pmatrix} 0 \\ -1 \\ 3 \end{pmatrix} + \frac{40}{35} \cdot \begin{pmatrix} -\frac{1}{5} \\ \frac{2}{5} \\ \frac{1}{5} \end{pmatrix}$$

$$u = \frac{3}{5} \cdot \begin{pmatrix} 0 \\ -1 \\ 3 \end{pmatrix} + \frac{8}{7} \cdot \begin{pmatrix} -\frac{1}{5} \\ \frac{2}{5} \\ \frac{1}{5} \end{pmatrix}$$

$$u = \begin{pmatrix} 0 \\ -\frac{3}{5} \\ \frac{9}{5} \end{pmatrix} + \begin{pmatrix} -\frac{8}{35} \\ \frac{16}{35} \\ \frac{8}{35} \end{pmatrix}$$

$$u = \begin{pmatrix} 0 \\ -\frac{21}{35} \\ \frac{63}{35} \end{pmatrix} + \begin{pmatrix} -\frac{8}{35} \\ \frac{16}{35} \\ \frac{8}{35} \end{pmatrix} = \begin{pmatrix} -\frac{8}{35} \\ -\frac{5}{35} \\ \frac{71}{35} \end{pmatrix}$$

$$c) \quad \|v - u\|^2 = \|v\|^2 - \|u\|^2$$

$$\|v\| = \sqrt{4+4} = \sqrt{8}$$

$$\|v\|^2 = 8$$

$$\|u\| = \sqrt{\left(\frac{8}{7}\right)^2 + \left(\frac{45}{35}\right)^2 + \left(\frac{55}{35}\right)^2} = \sqrt{\frac{38}{7}}$$

$$\|u\|^2 = \frac{38}{7}$$

$$\|v-u\|^2 = 8 - \frac{38}{7} = \frac{18}{7}$$

$$\|v-u\| = \sqrt{\frac{18}{7}}$$

$$\textcircled{7} \quad A^T \cdot A = I$$

$$A = \begin{pmatrix} \frac{1}{2}\sqrt{3} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2}\sqrt{3} \end{pmatrix}$$

$$A^T = \begin{pmatrix} \frac{1}{2}\sqrt{3} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2}\sqrt{3} \end{pmatrix}$$

$$\begin{pmatrix} \frac{1}{2}\sqrt{3} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2}\sqrt{3} \end{pmatrix} \cdot \begin{pmatrix} \frac{1}{2}\sqrt{3} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2}\sqrt{3} \end{pmatrix} = \begin{pmatrix} \frac{1}{4}\sqrt{3} \cdot \frac{1}{2}\sqrt{3} + \frac{1}{2} \cdot \frac{1}{2} & -\frac{1}{4}\sqrt{3} + \frac{1}{2}\sqrt{3} \\ \frac{1}{4}\sqrt{3} + \frac{1}{4}\sqrt{3} & -\frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2}\sqrt{3} \cdot \frac{1}{2}\sqrt{3} \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Quadratische Formen

③

$$Q(x, y, z) = -3x^2 - 2yx - 6zx - 5y^2 - 8yz - 5z^2$$

$$(x \ y \ z) \begin{pmatrix} -3 & -1 & -3 \\ -1 & -5 & -4 \\ -3 & -4 & -5 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$a = -3, b = -2, c = -6, d = -5, e = -8, f = -5$$

$$\left(\begin{array}{ccc|cc} -3-\lambda & -1 & -3 & -3-\lambda & -1 \\ -1 & -5-\lambda & -4 & -1 & -5-\lambda \\ -3 & -4 & -5-\lambda & -3 & -4 \end{array} \right)$$

det:

$$(-3-\lambda) \cdot (-5-\lambda) \cdot (-5-\lambda) + (-1 \cdot -4 \cdot -3) + (-3 \cdot -1 \cdot -4) - (9 \cdot (-5-\lambda)) - (16 \cdot (-3-\lambda)) - (-5-\lambda) = 0$$

$$(-3-\lambda) \cdot (-5-\lambda) \cdot (-5-\lambda) - 24 - (-45 - 45\lambda) - (-48 - 16\lambda) + 5 + \lambda = 0$$

$$(15 + 8\lambda + \lambda^2) \cdot (-5 - \lambda) + 21 + 45\lambda + 48 + 16\lambda + 5 + \lambda = 0$$
$$-75 - 40\lambda - 5\lambda^2 - 15\lambda - 8\lambda^2 - \lambda^3 + 21 + 45\lambda + 16\lambda + 5 + \lambda = 0$$
$$-37 + 7\lambda - 13\lambda^2 - \lambda^3 = 0$$

Quadriken

$$\textcircled{1} \quad A = \begin{pmatrix} \frac{16}{5} & \frac{2}{5} \\ \frac{2}{5} & \frac{19}{5} \end{pmatrix}$$

$$(A - \lambda I) = \begin{pmatrix} \frac{16}{5} - \lambda & \frac{2}{5} \\ \frac{2}{5} & \frac{19}{5} - \lambda \end{pmatrix}$$

$$\det(A - \lambda I) = \left(\frac{16}{5} - \lambda\right) \cdot \left(\frac{19}{5} - \lambda\right) - \left(\frac{2}{5} \cdot \frac{2}{5}\right) = 0$$

$$\frac{304}{25} - \frac{35}{5}\lambda + \lambda^2 - \frac{4}{25} = 0$$

$$\frac{300}{25} - \frac{35}{5}\lambda + \lambda^2 = 0$$

$$\lambda^2 - 7\lambda + 12 = 0$$

$$\lambda_{1,2} = \frac{7}{2} \pm \sqrt{\frac{49}{4} - \frac{48}{4}}$$

$$\lambda_{1,2} = \frac{7}{2} \pm \frac{1}{2}$$

$$\lambda_1 = 3, \lambda_2 = 4$$

$$p(\lambda) = (\lambda - 3) \cdot (\lambda - 4)$$

1. Probeprüfung

⑤ a) $x = \begin{pmatrix} 1 \\ 3 \\ 3 \end{pmatrix}, y = \begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix}$

$$\begin{pmatrix} 1 \\ 3 \\ 3 \end{pmatrix} - \begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -3 \\ 2 \\ 2 \end{pmatrix}$$

$$\|x - y\| = \sqrt{9 + 4 + 4}$$

$$\|x - y\| = \sqrt{17}$$

$$\cos \alpha = \frac{\langle x, y \rangle}{\|x\| \|y\|}$$

$$\langle x, y \rangle = \begin{pmatrix} 1 \\ 3 \\ 3 \end{pmatrix} \begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix} = 4 + 3 + 3 = 10$$

$$\alpha = \arccos \left(\frac{10}{\sqrt{19} \cdot \sqrt{18}} \right)$$

b) $\langle p, q \rangle = \int_{-1}^1 p(x) q(x) dx$

$$p(x) = x^2, q(x) = x^3 + 2x$$

$$\|p\| = \sqrt{\langle p, p \rangle} = \sqrt{\frac{2}{5}}$$

$$\int_{-1}^1 x^4 dx = \frac{x^5}{5} \Big|_{-1}^1 = \frac{1}{5} + \frac{1}{5} = \frac{2}{5}$$

$$\|q\| = \sqrt{\langle q, q \rangle} = \sqrt{\quad}$$

$$\int_{-1}^1 (2x + x^3) \cdot (2x + x^3) dx = \int_{-1}^1 4x^2 + 4x^4 + x^6 dx$$

$$= \left. \frac{x^3}{3} + \frac{4}{5}x^5 + \frac{1}{7}x^7 \right|_{-1}^1 =$$

$$\frac{1}{3} + \frac{4}{5} + \frac{1}{7} - \left(\frac{(-1)^3}{3} + \frac{4}{5} \cdot (-1)^5 + \frac{1}{7} \cdot (-1)^7 \right) =$$

$$\textcircled{6} \quad \lambda = -1, n = 2$$

$$\begin{pmatrix} 1 & 0 & 1 \\ 2 & 1 & -2 \\ 5 & 1 & 1 \end{pmatrix} \vec{v} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\text{I: } v_1 + v_3 = 0 \quad \Rightarrow v_1 = -v_3$$

$$\text{II: } 2v_1 + v_2 - 2v_3 = 0$$

$$\text{III: } 5v_1 + v_2 + v_3 = 0$$

$$\text{II: } -2v_3 + v_2 - 2v_3 = 0$$

$$v_2 = 4v_3$$

$$\text{IV: } -5v_3 + 4v_3 + v_3 = 0$$

$$v_3 = s, v_1 = -s, v_2 = 4s$$

$$\vec{v} = \begin{pmatrix} -1 \\ 4 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 1 \\ 2 & 1 & -2 \\ 5 & 1 & 1 \end{pmatrix} \cdot \vec{h} = \begin{pmatrix} -1 \\ 4 \\ 1 \end{pmatrix}$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & -1 \\ 2 & 1 & -2 & 4 \\ 5 & 1 & 1 & 1 \end{array} \right) \xrightarrow{r_2 - 2r_1} \left(\begin{array}{ccc|c} 1 & 0 & 1 & -1 \\ 0 & 1 & -4 & 6 \\ 5 & 1 & 1 & 1 \end{array} \right)$$

$$\text{I: } h_1 + h_3 = -1 \Rightarrow h_3 = -1 - h_1$$

$$\text{II: } h_2 - 4h_3 = 6 \Rightarrow h_2 + 4 + 4h_1 = 6$$

$$\text{III: } 5h_1 + h_2 + h_3 = 1$$

$$\text{II: } h_2 = 2 - 4h_1$$

$$\text{III: } 5h_1 + 2 - 4h_1 - 1 - h_1 = 1$$

$$h_1 = t, h_2 = 2 - 4t$$

$$\textcircled{11} \quad y''(t) - 2y'(t) - 8y(t) = 0$$

$$\lambda^2 - 2\lambda - 8 = 0$$

$$\lambda_{1,2} = 1 \pm \sqrt{1+8}$$

$$\lambda_{1,2} = 1 \pm 3$$

$$\lambda_1 = 4, \lambda_2 = -2$$

$$a) \{e^{4t}, e^{-2t}\}$$

$$b) \text{ Ansatz: } y_p = a e^{4t}$$
$$\dot{y} = 4a e^{4t}$$
$$\ddot{y} = 16a e^{4t}$$